

Standardization of NaOH Titrant and Volumetric Determination of Unknown Acid

Prelab reading: Harris, *Quantitative Chemical Analysis: Chapter 9*

1. Learning Goals

- Review the basics of titrations and titration calculations
- Understand the meaning of a primary standard
- Determine the precision of your analysis
- Verify your quantitative laboratory skills

2. Reagents

- Phenolphthalein indicator solution
- Unknown acid

3. Procedures

Note that this experiment will be carried out over two lab periods. In the first session, you will perform the standardization of NaOH. In the second session, you will titrate an unknown diprotic acid. In addition, you will be graded based on the quality of your data. If you do a poor job on your standardization of NaOH, you will not get good results for the diprotic acid titration.

Titration of the unknown acid (week 2)

- Obtain ~70 mL of your unknown acid. This will be enough for both titrations.
- Pipette 25 mL of the unknown acid into an Erlenmeyer flask and add a small stir bar to the solution. Stir bars are kept near the front of the lab in the grey tackle box. Add 3 drops of indicator to the unknown acid solution and stir.

Calibrate your pH electrode

- pH electrodes are prone to drift and need to be calibrated often.
- Use pH 4 and 10 buffer solutions to calibrate the pH electrodes. Use the calibration menu on the lab quest unit and follow the directions from this menu.
- Ask instructor for help if you are having trouble.

Start of your preliminary run – to find out where your equivalence volume occurs.

- Place your pH electrode in the beaker in such a way that the stir bar will not hit the bottom of the electrode and start the stir bar with gentle stirring. Carefully move your ring stand such that the burette will drain into your flask. Make sure that the entire glass bulb on the underside of the pH electrode is submerged in solution.
- Read your pH meter. Record pH and your V_i on your burette.
- Add ~1 mL of NaOH. Record pH and your V on your burette. Repeat this step until the pH of the solution is >11.5
- From your pH and volume data, estimate your two equivalence points to the nearest 1 mL. It is helpful if you have a laptop to do this, but you can also do it by looking at the difference in pH for each 1-mL increment.

Start of your accurate titration – Prepare a fresh sample of the unknown diprotic acid using the procedure above.

- Add ~1 mL of NaOH. Record volume and pH.

- Repeat until you are within 1.5 pH units of your equivalence point. At this point add your NaOH in ~ 0.2 mL aliquots. Record volume and pH with each addition. Repeat with the small steps until your pH is essentially unchanged with the addition of the NaOH solution.
- Add ~1 mL aliquots of NaOH until the pH is >11.5.

4. Cleanup

- Clean and rinse all the glassware that you have used and place it on the drying rack. Place volumetric pipettes in the oven (careful: oven is hot!).
- Wipe your bench space clean.
- Dispose of all other wastes in the designated waste container.

5. Data Analysis and Presentation of Results in Lab Reports

NaOH standardization

- For each standardization titration, calculate the molarity of the NaOH titrant.
- Express the NaOH concentrations to at least four decimal places (e.g., 0.1023 M, or 0.0997 M).
- Calculate the average value, standard deviation, and RSD of your NaOH molarities.
- Be sure to report your final result along with the precision (95% confidence interval) in the form of $Interval = \bar{x} \pm \frac{ts}{\sqrt{n}}$. You will be graded mainly on the accuracy of your result. When you get t from the *Student's t* table (Table 4-2), use $n-1$ as the degrees of freedom, where n is the number of measurements.
- If one result is quite different from the others, and there is no experimental reason to discard it, you may apply the Q test (at least 4 titrations needed for Q test). If this test rejects one of the molarity values, average the remaining values. It is always better to have more data; therefore run additional titrations as needed to obtain precise and accurate results.

Run	Volume of NaOH	Corrected Volume of NaOH	Concentration of NaOH	Average	SD	RSD
1						
2						
3						
4						
...						

Unknown acid titration.

- Plot your pH vs volume of NaOH data in Excel.
- Calculate the first and second derivatives using Excel.
- Plot the second derivative versus volume of NaOH and determine the 1st and 2nd equivalence volumes.
- From your equivalence volumes determine your 1/2 equivalence volume. Determine the pH at this volume.
- Determine the K_a for your unknown diprotic acid.
- Using the calculated K_a values and the tabulated data for K_a in your textbook (Appendix G in the 9th Ed.), determine the identity of the unknown diprotic acid.

- Determine the concentration of your unknown diprotic acid for each titration.

6. Questions

1. Why is KHP used as a primary standard material?
2. At the endpoint, the solution will be faintly pink, and the color can stable for at about 20 seconds, but it will eventually fade. Why?
3. Standardized NaOH solutions are typically re-standardized if left unused for more than a week or two. The main concern is exposure to the atmosphere (thus it is important to seal the bottle well each time and not leave it open during a lab period). Why is exposure to the atmosphere a concern (be specific)?
4. Why do we need to perform the blank titration?
5. To standardize a NaOH solution, a titration net volume should be 20.00 mL and a student accidentally goes one drop past this endpoint (~ 0.05 mL), what will the relative error in this titration be? Will the calculated [NaOH] be higher or lower than the true value of that solution's concentration? Explain briefly.
6. We can measure the concentration of NaOH solution by reaction with KHP. Complete reaction with 0.9674 ± 0.0009 g of KHP (molecular weight: 204.2210 ± 0.0007 g/mol) required 27.35 ± 0.04 mL of NaOH. Find the molarity of the NaOH and its absolute uncertainty.

Standardization of the pH electrode

Keep electrodes in aqueous solution when not in use.

The glass - bubble bottom of the electrode is very fragile. Do not touch or hit with stir bar during experiment.

1. Rinse electrode with DI water. Place in pH 7 standard solution (green-yellow)
2. Turn meter **on**
3. if the display doesn't show "pH" in upper left corner, press **pH** until it does
4. Press **calibrate** key
5. display should read 7.00. Press **Enter**
6. The 7.00 in the display will start to flash. May take a couple of minutes, but it should stop flashing and then the display will read 4.01 with 2 PT in the middle right.
7. Rinse electrode with more DI water. Place in pH 4 standard solution (pink)
8. Press **Enter**
9. The 4.01 in the display will start to flash. May take a couple of minutes, but it should stop flashing and then the display will read 4.00 ish – means the meter is calibrated. Don't turn off meter until class is over
10. If the pH display only reads to 0.1 pH unit, press the pH key twice – should now read to nearest 0.01 pH unit.
10. Rinse electrode and place back into water until ready for first reading.

To get pH reading – rinse off electrode with water. Place in solution. Don't record the number in the display until the "pH" in the upper left corner stops flashing (means you have a stable reading).